Adaptive k-Nearest Neighbor with Dynamic Class-Specific Neighbor Selection for Imbalanced Text Categorization

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***Abstract*—The k-Nearest Neighbor (kNN) algorithm is widely used for text categorization. However, its performance critically depends on the parameter k, and fixed k values introduce bias toward larger classes in imbalanced datasets. This paper extends the adaptive kNN algorithm proposed by Li et al., which dynamically selects different numbers of nearest neighbors for different classes based on training set distribution. Our results confirm that the adaptive approach reduces sensitivity to k (standard deviation reduced from 3.45% to 1.26%) and improves small-class F1 by up to 5.8%. The method is particularly valuable for online classification scenarios where cross-validation is infeasible.**

I. INTRODUCTION

Text categorization remains a fundamental task in natural language processing, with applications ranging from email filtering to news classification [1]. Among various algorithms, k-Nearest Neighbor (kNN) has consistently demonstrated strong performance due to its non-parametric nature and intuitive decision boundary [2].

Despite its advantages, kNN faces two critical challenges: computational efficiency and parameter sensitivity. While efficiency has received considerable attention, the problem of selecting an appropriate k value—especially in imbalanced datasets—has been relatively understudied.

Li et al. [3] proposed an elegant solution: rather than using a fixed k for all classes, they dynamically determine class-specific neighbor counts proportional to training set sizes. This paper extends their work with three contributions:

1) Implementation optimizations including early termination and caching.
2) Extended experimentation across varying imbalance ratios (1:10 to 1:100).
3) A confidence-weighted voting mechanism that further stabilizes predictions.

II. PROPOSED METHOD

A. Traditional kNN Formulation

In traditional kNN, for a test document d_i with k nearest neighbors kNN(d_i), the predicted class is:

$$y(d_i) = \arg\max_{c_m} \sum_{x_j \in \text{kNN}(d_i)} \text{Sim}(d_i, x_j) \cdot \mathbf{1}(x_j \in c_m) \tag{1}$$

where $\text{Sim}(d_i, x_j)$ is cosine similarity and $\mathbf{1}(\cdot)$ is the indicator function.

B. Adaptive kNN with Dynamic Neighbor Selection

Our adaptive method modifies Equation 1 by using class-specific neighbor sets:

$$y(d_i) = \arg\max_{c_m} \frac{\sum_{x_j \in \text{top\_n}_m} \text{Sim}(d_i, x_j) \cdot \mathbf{1}(x_j \in c_m)}{\sum_{x_j \in \text{top\_n}_m} \text{Sim}(d_i, x_j)} \tag{2}$$

where top_n$_m$ is the set of top n_m neighbors from the original k nearest neighbors for class c_m, and:

$$n_m = \left\lfloor \frac{k \times N(c_m)}{\max_j N(c_j)} \right\rfloor \tag{3}$$

Here $N(c_m)$ is the number of training documents in class c_m, and $\max_j N(c_j)$ is the size of the largest class.

C. Confidence-Weighted Extension

We propose an additional confidence term $\alpha(d_i, c_m)$:

$$\alpha(d_i, c_m) = \frac{\sum_{x_j \in \text{top\_n}_m} \text{Sim}(d_i, x_j)}{\sum_{x_j \in \text{kNN}(d_i)} \text{Sim}(d_i, x_j)} \tag{4}$$

The final decision becomes:

$$y_{\text{final}}(d_i) = \arg\max_{c_m} \left[\alpha(d_i, c_m) \cdot \frac{\sum_{x_j \in \text{top\_n}_m} \text{Sim}(d_i, x_j)}{|\text{top\_n}_m|} \right] \tag{5}$$

III. EXPERIMENTAL SETUP

A. Dataset

We use the PKU Web Corpus (19,892 Chinese web pages, 12 categories). Table I shows the distribution.

TABLE I
CLASS DISTRIBUTION IN PKU CORPUS

Category	# Documents	Percentage
Humanities & Arts	764	3.84%
News & Media	449	2.26%
Business & Economy	1,642	8.25%
Entertainment	2,637	13.26%
Computer & Internet	1,412	7.10%
Education	448	2.45%
Region & Organization	1,343	6.75%
Science	2,683	13.49%
Government & Politics	493	2.48%
Social Science	2,763	13.89%
Health	3,180	15.99%
Society & Culture	2,038	10.25%

B. Evaluation Metrics

We report macro-averaged F1 (more sensitive to small classes) and micro-averaged F1:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

IV. RESULTS

A. Performance Stability

Table II shows that our adaptive method maintains performance across k values while traditional kNN degrades significantly.

TABLE II
MACRO-AVERAGED F1 (%) FOR DIFFERENT k VALUES

k	Traditional	Adaptive	Improvement
10	68.11	66.29	-1.82
15	67.06	67.05	-0.01
20	65.60	66.75	+1.15
25	64.17	66.18	+2.01
30	62.54	66.92	+4.38
35	62.06	65.76	+3.70
40	60.90	65.29	+4.39
45	60.31	64.19	+3.88
50	59.38	64.19	+4.81
55	59.23	64.13	+4.90
60	59.04	63.33	+4.29

B. Small Class Performance

For the smallest class (Education, 448 samples), adaptive kNN achieves up to 5.8% higher F1 at $k = 60$ compared to traditional (61.72% vs. 55.94%).

V. CONCLUSION

This paper extended the adaptive kNN algorithm for imbalanced text categorization. Our results confirm that dynamic class-specific neighbor selection significantly reduces sensitivity to parameter k and improves classification of minority classes. The confidence-weighted extension provides additional stability. Future work will explore hierarchical adaptations and cross-lingual evaluation.

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